Differentiation

Week 3A

Overview

Introduction to Differentiation

Key Ideas - The Gradient Function

Techniques of Derivatives

Key Ideas - Derivative of Powers of x

- Rule 1: Constant Multiplier Rule

- Rule 2: Sum Rule

Key Idea – Gradient Function

Differentiation is a technique used to calculate the **gradient, or slope, at any point on a graph**.

The Gradient Function

Given a function, for example $y = x^2$, it is possible to **derive a formula for the gradient of its graph.** We can think of this formula as the **gradient function,** because it allows us to find the gradient of the graph at any point.

For example, when $y = x^2$, the gradient function is $2x$.

We will firstly look at the fundamental rule for finding the gradient function but will not derive it from first principles. Which is a more detailed method of finding the gradient function. The important point here is that we can use this formula to calculate the **gradient of a functions at different poin**ts on the graph.

For the above example,

when $x = 3$, the gradient is $2 \times 3 = 6$

when $x = -2$, the gradient is $2 \times (-2) = -4$

Interpreting the Gradient

A gradient of 6 means that y is increasing at a rate of 6 units for every 1 unit increase in x.

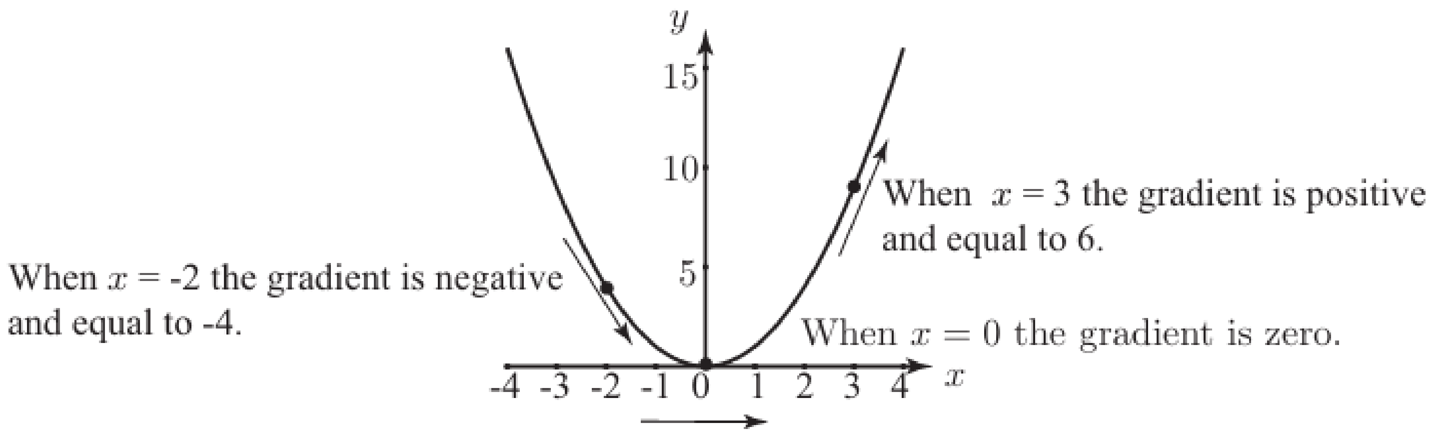
A gradient of −4 means that y is decreasing at a rate of 4 units for every 1 unit increase in x.

Note that when $x = 0$ the gradient is $2 \times 0 = 0$.

If you look at the graph of $y = x^2$

- At $x = 3$, the graph has a positive gradient
- At $x = -2$, the graph has a negative gradient
- At $x = 0$, the gradient is zero

These features of the graph can be obtained directly from the gradient function $2x$



Notation for the Gradient Function

You will need to use correct notation for the **gradient function**. If y is a function of x, that is $y = f(x)$, we write its **gradient function as** $\boldsymbol{\frac{dy}{dx}}$.

It is pronounced "dee y by dee x" but is not a fraction, even though it looks like one.

The process of finding $\frac{dy}{dx}$ is called **differentiation with respect to *x***.

The gradient function of the power function $f\left(x\right)=x^n$, where n is a real number, is given by

$$\boldsymbol{\frac{dy}{dx}=nx^{n-1}}$$

Note: The gradient function can be found from first principles, but we are NOT going to explain it in this subject.

More Notation and Terminology

When $y = f(x)$, the gradient function can be written as:

$$\frac{dy}{dx}, \quad y' \quad \text{or} \quad f' \quad \text{or} \quad \frac{d}{dx}f(x)$$

y' (pronounced "y dash") or f' (pronounced "f dash").

A **derivative** is another name for the gradient function.

It is also commonly described as the **rate of change** of a function.

Summary of different ways questions can be asked:

Common wording :

- Find the derivative
- Differentiate the function
- Find the gradient function
- Find the rate of change
- Find the first derivative

Common notation:

- Find/Determine $f'(x)$ or y' or $\dfrac{dy}{dx}$ or $\dfrac{d}{dx}f(x)$

Wording at a point $x = a$:

- Find the gradient at $x = a$
- Find $f'(a)$
- Find the slope of the tangent at $x = a$
- Find the gradient of the curve at a $x = a$
- Find the derivative at $x = a$

Conceptual / Word-Based

- Find the rate of change of the function.
- Determine how fast the function is changing.
- Find the instantaneous rate of change.
- Determine the slope of the curve at a given point

Key Idea - Derivatives of Powers of x

The derivative of the power function $f(x) = x^n$, where n is a real number, is given by

$$\frac{dy}{dx} = nx^{n-1}$$

Note: Rewrite reciprocal indices as negative indices.

If the variable is in the denominator, express it with a negative index then differentiate.

$$\boldsymbol{\frac{1}{a^n} = a^{-n}}$$

For example, , $\frac{1}{a^4} = a^{-4}$

***Examples*:** Differentiate the following, and write your answers with positive indices:

(a) $y = x^5$ $\qquad \frac{dy}{dx} = 5x^4$ or $f'(x) = 5x^4$

(b) $y = \frac{1}{x^7} = x^{-7}$ $\qquad \frac{dy}{dx} = -7x^{-8}$

$$= \frac{-7}{x^8}$$

Exercices:

Find the derivative of the following functions, write your answer in positive index form:

(a) $y = x^7$

$$\frac{dy}{dx} = 7x^{7-1} = 7x^6$$

(b) $f(x) = \dfrac{1}{x^6}$ $= x^{-6}$

$$\frac{dy}{dx} = -6x^{-6-1} = -6x^{-7} = \frac{-6}{x^7}$$

Discussion Question

What is the derivative of $y = x$ and $y = 8$?

$y = x$ can also be written as $y = x^1$

$y' = 1x^{1-1} = 1x^0 = 1$

$y = 8 \Rightarrow y' = 0$

What can you conclude about the derivative of any linear function $f(x) = mx + c$?

$f'(x) = m$

What can you conclude about the derivative of any constant $f(x) = c$?

$f'(x) = 0$

Be Careful with Setting out and Notation!!!!!!

For example, if the question asked you to differentiate x^4, any of the following notations are acceptable:

- $\frac{dy}{dx} = 4x^3$
- $y' = 4x^3$
- $f'(x) = 4x^3$

Do NOT write $x^4 = 4x^3$ as this is INCORRECT

We know that the rule for finding the derivative of $y = x^n$ applies to all real values of n. However in all of our examples so far, n has been an integer.

We will now consider cases where ***n* is a fraction** (a rational number)

When the power n is a fraction, it **may be necessary to rearrange** the function before differentiating. In particular, there are three important cases to remember:

1. Rewrite surds (roots) as indices.

Expressions involving surds can be written with fractional indices using:

$$\boldsymbol{a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m}$$

For example, $\sqrt{x} = x^{1/2}$, $\sqrt[3]{x^2} = x^{2/3}$

2. Rewrite reciprocal surds as negative indices.

If the variable is in the denominator, express it with a negative index

$$\boldsymbol{\frac{1}{a^{m/n}} = a^{-m/n} \quad , \quad \frac{1}{\sqrt[n]{a^m}} = a^{-m/n}}$$

For example, $\dfrac{1}{x^{2/5}} = x^{-2/5}$, $\dfrac{2}{\sqrt[3]{x}} = 3x^{-1/3}$

3. Apply index laws before differentiating (multiplication/division).

When expressions involve multiplication or division of powers, simplify first using index laws:

$$\boldsymbol{a^n \times a^m = a^{n+m} \quad a^n \div a^m = a^{n-m}}$$

For example, $a^4 \times a^2 = a^6, \quad a^5 \div a^2 = a^3$

Note: You need to understand the difference between leaving your answer in positive index form and surd form and you must be able to write both forms. In the exam, the question will tell you which form to use for your final answer.

For example,

- $x^{-\frac{1}{3}}$ is in negative index form
- $\dfrac{1}{x^{\frac{1}{3}}}$ is in positive index form
- $\dfrac{1}{\sqrt[3]{x}}$ is in surd form

Remember: For example, $\dfrac{1}{3}x^{\frac{2}{3}} = \dfrac{x^{\frac{2}{3}}}{3}$ and $\dfrac{3}{5}x^2 = \dfrac{3x^2}{5}$

Examples: Differentiate the following, writing your answer in positive index form and in surd form:

(a) $y = x^{4/5}$

$$\frac{dy}{dx} = \frac{4}{5}x^{-1/5}$$

$$= \frac{4}{5x^{1/5}} \text{ (writing the answer with a positive index)}$$

$$= \frac{4}{5\sqrt[5]{x}} \text{ (writing the answer with a surd)}$$

(b) $f(x) = x^{\frac{2}{3}}$

$$f'(x) = \frac{2}{3}x^{-\frac{1}{3}} = \frac{2x^{-\frac{1}{3}}}{3}$$

$$= \frac{2}{3x^{\frac{1}{3}}} \text{ (positive index form)}$$

$$= \frac{2}{3\sqrt[3]{x}} \text{ (surd form)}$$

(c) $y = -\sqrt[3]{x^2}$

Firstly, we need to rewrite the expression in index form.

$$y = -\sqrt[3]{x^2} = -x^{2/3} \qquad \frac{dy}{dx} = -\frac{2}{3}x^{-1/3} = -\frac{2x^{-1/3}}{3}$$

$$= \frac{-2}{3x^{1/3}} \text{ (positive index form)}$$

$$= \frac{-2}{3\sqrt[3]{x}} \text{ (surd form)}$$

(d) $y = \dfrac{1}{x^{1/5}}$

Firstly, we need to rewrite the expression in index form.

$$y = \frac{1}{x^{1/5}} = x^{-1/5} \qquad \frac{dy}{dx} = -\frac{1}{5}x^{-6/5}$$

$$= \frac{1}{5x^{6/5}} \text{ (positive index form)}$$

$$= \frac{1}{5\sqrt[5]{x^6}} \text{ (surd form)}$$

(e) $f(x)=\dfrac{1}{\sqrt[3]{x}}$

Firstly, we need to rewrite the expression in negative index form.

$$f(x)=\frac{1}{\sqrt[3]{x}}=x^{-1/3} \qquad f'(x)=-\frac{1}{3}x^{-4/3}=-\frac{x^{-4/3}}{3}$$

$$=-\frac{1}{3x^{4/3}} \text{ or } -\frac{1}{3\sqrt[3]{x^4}} \text{ (surd form)}$$

(f) $f(x)=x\sqrt[3]{x}$

Firstly, simplify using index laws:

$$f(x)=x\sqrt[3]{x}=x\times x^{\frac{1}{3}}=x^{\frac{4}{3}} \qquad f'(x)=\frac{4x^{1/3}}{3} \text{ or } \frac{4\sqrt[3]{x}}{3}$$

(g) $y=\dfrac{\sqrt{x}}{\sqrt[3]{x}}$

Firstly, simplify using index laws:

$$y=\frac{\sqrt{x}}{\sqrt[3]{x}}=\frac{x^{1/2}}{x^{1/3}}=x^{1/2-1/3}=x^{1/6} \qquad y'=\frac{1}{6}x^{-5/6}$$

$$=\frac{1}{6x^{5/6}} \text{ or } \frac{1}{6\sqrt[6]{x^5}}$$

Exercices:

Find the gradient function of the following. Write your answer in positive index form:

only

(a) $y = \sqrt{x}$ $= x^{\frac{1}{2}}$

$y' = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2x^{\frac{1}{2}}}$

in this question you donot need to write the answer in surd form

(b) $y = \frac{1}{x^{\frac{1}{3}}}$ $= x^{-\frac{1}{3}}$

$\frac{dy}{dx} = -\frac{1}{3}x^{-\frac{1}{3}-1} = -\frac{1}{3}x^{-4/3} = \frac{-1}{3x^{4/3}}$

(c) $y = x^{2.5}$

$y' = 2.5x^{2.5-1} = 2.5x^{1.5}$

(d) $y = \frac{1}{\sqrt[3]{x^5}}$ $= x^{-5/3}$

$y' = -\frac{5}{3}x^{-5/3-1} = \frac{-5}{3}x^{-8/3} = \frac{-5}{3x^{8/3}}$

Exercices:

Find the derivative of the following functions, writing your answer in positive index form:

(a) $f(x)=\sqrt{x}\times x^{3}$ =

$= x^{\frac{1}{2}}\times x^{3}$

$= x^{\frac{1}{2}+3}$

$= x^{7/2}$

$f'(x) = \frac{7}{2}x^{7/2-1} = \frac{7}{2}x^{5/2}$

(b) $y=\sqrt{x}\div x^{1/3}$

$= x^{\frac{1}{2}}\div x^{1/3}$

$= x^{\frac{1}{2}-\frac{1}{3}}$

$= x^{\frac{1}{6}}$

$y' = \frac{1}{6}x^{\frac{1}{6}-1} = \frac{1}{6}x^{-5/6} = \frac{1}{6x^{5/6}}$

Key Idea – Rule 1 and Rule 2

There are five fundamental rules of differentiation that apply to all differentiable functions, including logarithmic, trigonometric, and exponential functions. These rules are essential, as they form the basis for differentiating more complex expressions.

Before we continue with further derivatives of the power function, we will introduce the first two of these basic rules then move on to the last 3 rules.

Rule 1 - Constant multiplier rule

If $y = f(x)$ and k is a constant, then

$$\frac{d}{dx}(kf) = kf'(x) = k\frac{dy}{dx}$$

***Examples*:** Use Rule 1 to differentiation the following, writing your answer in positive index form and in surd form:

(a) $y = 3x^7$

$$y' = 3 \times 7x^6$$
$$= 21x^6$$

(b) $f(x) = \dfrac{3}{x^6} = 3x^{-6}$

$$f'(x) = 3x^{-6}$$
$$= 3 \times (-6)x^{-7}$$
$$= -18x^{-7}$$
$$= \frac{-18}{x^7}$$

(c) $y = -3\sqrt{x} = -3x^{\frac{1}{2}}$

$$\frac{dy}{dx} = -3 \times \frac{1}{2} x^{-\frac{1}{2}}$$
$$= -\frac{3}{2} x^{-\frac{1}{2}} \quad \left(\text{or} \quad -\frac{3x^{-\frac{1}{2}}}{2} \right)$$
$$= -\frac{3}{2x^{\frac{1}{2}}} \quad \text{positive index form}$$
$$\text{or} \quad -\frac{3}{2\sqrt{x}} \quad \text{surd form}$$

Rule 2 - Sum rule

If $u = f\left(x\right)$ and $v = g(x)$ are two functions, then

$$\boldsymbol{\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx} = f'(x) \pm g'(x)}$$

***Examples*:** Using Rule 2 to differentiation the following:

(a) $y = x^5 + 2x^3 - 4x + 8$

$$\frac{dy}{dx} = 5x^4 + 6x^2 - 4$$

(b) $y = 2x^{-3} - \sqrt[3]{x}$

$$y' = -6x^{-4} - \frac{1}{3}x^{-\frac{2}{3}}$$

$$= \frac{-6}{x^4} - \frac{1}{3x^{\frac{2}{3}}}$$

$$= \frac{-6}{x^4} - \frac{1}{3\sqrt[3]{x^2}}$$

Exercise:

Using the first two differentiation rules, differentiate the following functions. Write your answers in surd form and in positive index form where necessary:

(a) $f(x) = x^3 - 2x - 4$

$f'(x) = 3x^{3-1} - 2 \times 1 x^{1-1} - 0$

$f'(x) = 3x^2 - 2$

(b) $f(x) = x^2 + 3\sqrt{x} - \dfrac{2}{x^3} + \dfrac{1}{\sqrt[4]{x}}$ $= x^2 + 3x^{\frac{1}{2}} - 2x^{-3} + x^{-\frac{1}{4}}$

$f'(x) = 2x^{2-1} + 3 \times \frac{1}{2} x^{\frac{1}{2}-1} - 2 \times (-3) x^{-3-1} + \left(-\frac{1}{4}\right) x^{-\frac{1}{4}-1}$

$= 2x + \frac{3}{2} x^{-\frac{1}{2}} + 6x^{-4} - \frac{1}{4} x^{-5/4}$

$f'(x) = 2x + \dfrac{3}{2\sqrt{x}} + \dfrac{6}{x^4} - \dfrac{1}{4\sqrt[4]{x^5}}$

(c) $f(x) = \dfrac{3x^2}{\sqrt{x}} + \dfrac{2x^5}{3}$ $= \dfrac{3x^2}{x^{1/2}} + \dfrac{2x^5}{3}$

$f(x) = 3x^{2-\frac{1}{2}} + \frac{2}{3}x^5 = 3x^{3/2} + \frac{2}{3}x^5$

$f'(x) = 3 \times \frac{3}{2} x^{\frac{3}{2}-1} + \frac{2}{3} \times 5 x^{5-1}$

$f'(x) =$

$f'(x) = \dfrac{9}{2} x^{\frac{1}{2}} + \dfrac{10}{3} x^4 = \dfrac{9\sqrt{x}}{2} + \dfrac{10x^4}{3}$

In many cases it is necessary to expand or simplify the function before differentiation using **Sum rule and Constant multiplier rule**

***Examples*:**

(a) Differentiate $f(x) = x(x-3)$

$$f(x) = x(x-3)$$
$$= x^2 - 3x$$
$$f'(x) = 2x - 3$$

(b) Differentiate $y = (3x+2)(2x+1)$

$$y = (3x+2)(2x+1)$$
$$= 6x^2 + 7x + 2$$
$$\frac{dy}{dx} = 12x + 7$$

(c) Differentiate $y = \dfrac{4x^3 - 2x}{x}$

$$y = \frac{4x^3 - 2x}{x} \text{ OR } x^{-1}(4x^3 - 2x)$$
$$= \frac{4x^3}{x} - \frac{2x}{x}$$
$$= 4x^2 - 2$$
$$y' = 8x$$

(d) Differentiate $f(x) = \dfrac{10x^2 + 15}{5x}$

$$f(x) = \frac{10x^2 + 15}{5x}$$
$$= \frac{10x^2}{5x} + \frac{15}{5x}$$
$$= 2x + \frac{3}{x} = 2x + 3x^{-1}$$
$$f'(x) = 2 + 3(-1)x^{-2} = 2 - \frac{3}{x^2}$$

Exercises:

(a) Find the derivative of $f(x)=(x+3)^2$. $= x^2+6x+9$

$$f'(x) = 2x^{2-1} + 6\times 1 x^{1-1} + 0$$

$$f'(x) = 2x+6$$

(b) Calculate $\frac{dy}{dx}$ for $y=\frac{6x^3+5}{2x}$ $= \frac{6x^3}{2x} + \frac{5}{2x}$

$$= 3x^2 + \frac{5}{2}x^{-1}$$

$$\frac{dy}{dx} = 3\times 2x^{2-1} + \frac{5}{2}(-1)x^{-1-1}$$

$$= 6x - \frac{5}{2}x^{-2} = 6x - \frac{5}{2x^2}$$